

CyberTech Solutions - Enterprise LAN Network Design

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1. Project Overview

Objective

This project demonstrates the design and implementation of a segmented enterprise Local Area Network (LAN) for a fictional company, CyberTech Solutions, using Cisco Packet Tracer.

The primary objective was to build a scalable, manageable, and secure network infrastructure by applying industry best practices, including VLAN segmentation, structured IP addressing, and Layer 3 switching.

Technologies

- Cisco Packet Tracer
- Cisco IOS
- Ethernet Switching
- IPv4
- VLSM (Variable Length Subnet Masking)
- VLANs
- Layer 3 Switching
- Inter-VLAN Routing

2. Company Requirements

CyberTech Solutions is a medium-sized IT company consisting of multiple departments with different networking requirements.

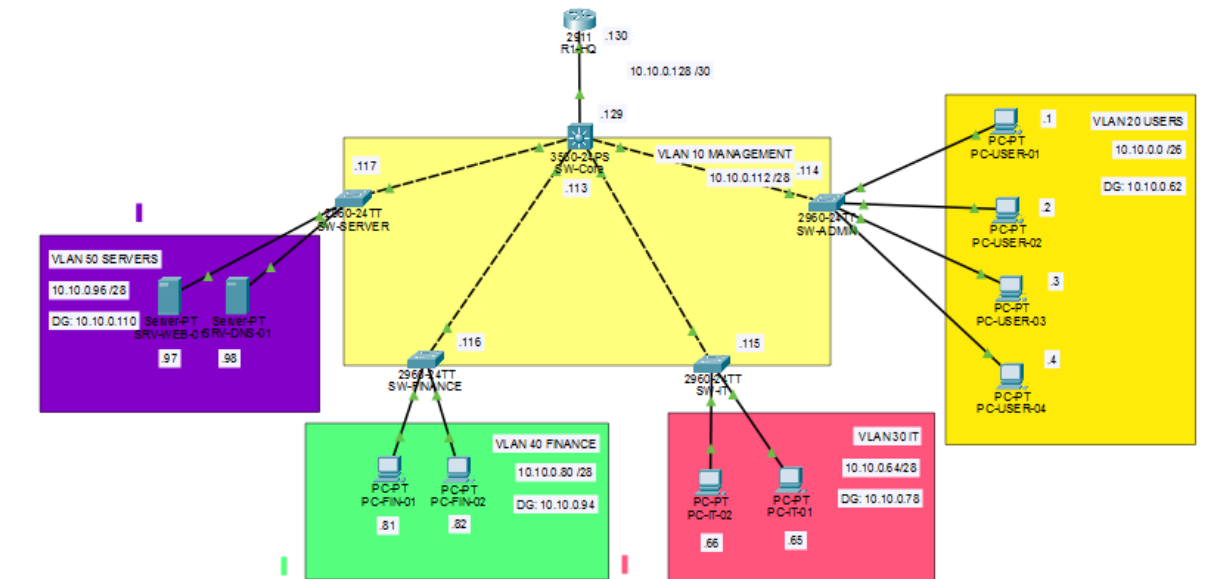
Departments

- Management
- Employees
- IT
- Finance
- Server Infrastructure

Network Requirements

- Separate departments into independent broadcast domains.
- Allow controlled communication between departments.
- Centralize routing using a Layer 3 switch.
- Provide a dedicated management network for network devices.
- Design an efficient IPv4 addressing scheme using VLSM.
- Create a scalable architecture that supports future growth.

3. Network Topology



The network follows a hierarchical design consisting of a Layer 3 core switch connected to multiple Layer 2 access switches. End-user devices are connected through dedicated access switches, while servers are placed in a separate server network.

4. Network Design Decisions

VLAN Segmentation

Each department was assigned its own VLAN to improve security, reduce broadcast traffic, and simplify network administration.

Layer 3 Core Switch

A Layer 3 switch was selected as the network core to perform inter-VLAN routing locally, reducing the routing workload on the edge router and improving overall network performance.

Management VLAN

A dedicated Management VLAN was implemented for administering network devices. This separates management traffic from user traffic and provides a more secure administration environment.

VLSM Addressing

Variable Length Subnet Masking (VLSM) was used to allocate address space according to the number of devices required in each department, minimizing wasted IP addresses while allowing room for future expansion.

Dedicated Server Network

Servers were placed in a separate VLAN to isolate critical infrastructure from end-user devices and prepare the network for future access control policies.

5. VLAN Plan

VLAN	Name	Purpose
10	Management	Network device management
20	Users	Employee workstations
30	IT	IT department
40	Finance	Finance department
50	Servers	Server infrastructure
60	<i>Corporate Wi-Fi (planned)</i>	Employee wireless access
70	<i>Guest Wi-Fi (planned)</i>	Visitor wireless access

6. IP Addressing Plan

VLAN	Network	Gateway	Subnet Mask
10	10.10.0.112		255.255.255.240
20	10.10.0.0	10.10.0.62	255.255.255.192
30	10.10.0.64	10.10.0.78	255.255.255.240
40	10.10.0.80	10.10.0.94	255.255.255.240
50	10.10.0.96	10.10.0.110	255.255.255.240

7. Implementation Summary

The following tasks were completed during the implementation:

- Created VLANs for all departments.
- Assigned access ports to the appropriate VLANs.
- Configured trunk links between the core and access switches.
- Implemented Switch Virtual Interfaces (SVIs).
- Enabled inter-VLAN routing on the Layer 3 switch.
- Assigned management IP addresses to Layer 2 switches.
- Configured default gateways for network management.
- Verified connectivity between VLANs.

8. Validation

The network was tested after implementation.

Validation included:

- Successful communication between hosts within the same VLAN.
- Successful inter-VLAN routing between departments.
- Connectivity verification using ICMP (ping).
- Verification of VLAN assignments and routing configuration using Cisco IOS commands.

9. Future Improvements

The following features are planned as future enhancements:

- Secure device administration using SSH
- DHCP services
- Wireless infrastructure
- Guest Wi-Fi isolation
- Access Control Lists (ACLs)
- Internet connectivity with NAT
- Dynamic routing (OSPF)
- Network monitoring and logging

10. Conclusion

This project demonstrates the design and implementation of a structured enterprise LAN using Cisco technologies. It applies networking fundamentals such as VLAN segmentation, VLSM addressing, Layer 3 switching, and inter-VLAN routing while following scalable network design principles. The resulting infrastructure provides a solid foundation for future security enhancements and enterprise network services.